

Business Problem

Commercial airlines operate in a highly competitive environment where profitability depends on making informed decisions regarding network planning, aircraft utilization, and flight scheduling. One of the most critical scheduling decisions is how often to operate a route; scheduling too many flights on routes with insufficient demand can increase operating costs and reduce profitability, while scheduling too few flights on high-demand routes may result in lost revenue and diminished customer satisfaction. As airline networks expand and market conditions continue to evolve, accurately forecasting route demand has become increasingly important. Traditionally, airlines have relied on historical schedules, market analyses, passenger demand trends, and industry expertise to develop future schedules. While these methods are valuable, the growing availability of large operational datasets has created new opportunities to enhance forecasting through predictive analytics and machine learning. These advanced techniques can uncover complex relationships between operational variables that may not be discovered through traditional statistical methods. This enables analysts to produce more accurate demand forecasts and support more effective strategic decision-making.

The objective of this project is to develop a machine learning model to predict the number of scheduled monthly departures for an airline route using historical airline scheduling data from the SABRE Market Intelligence database. By identifying the operational characteristics that strongly influence departure frequency, the model looks to provide insights that could support airline scheduling, fleet planning, and network development decisions. Although the model is intended as a decision-support tool rather than a replacement for human expertise, accurate demand forecasts can help airlines improve resource allocation and operational efficiency.

To accomplish this objective, the project follows a comprehensive applied data science workflow considering data preparation, exploration, analysis, predictive modeling, model evaluation, and interpretation of results. Attention is given to ensuring that the final model reflects realistic forecasting conditions by identifying and correcting potential sources of data leakage. This approach not only improves predictive reliability but also demonstrates the importance of rigorous model validation when developing machine learning solutions for businesses.

Table 1. Dataset Summary

Metric	Value
Original Records	250786
Original Variables	24
Final Modeling Records	250786
Final Modeling Variables	20
Training Records	200628
Testing Records	50158
Missing Values Removed	232

Table 1. Dataset Summary

Table 1 summarizes the size of the original dataset, the size of the final modeling dataset after preprocessing, the number of training and testing observations, and the records removed during data cleaning.

Data Explanation

Dataset Overview

The primary data source for this project was the SABRE Market Intelligence schedules database, which contains scheduled commercial airline operations throughout North America. The dataset represents airline scheduling activity for the 2026 and 2027 operating periods and includes detailed operational information for individual origin-destination markets. Each observation corresponds to a scheduled airline operating within a specific airport market during a particular month (SABRE Corporation, n.d.).

The original dataset contained 251,018 observations across 24 variables, providing a comprehensive view of airline scheduling activity. Variables included airport market identifiers, origin and destination airports, airline and fleet information, aircraft manufacturer, schedule type, monthly departure counts, airline capacity, available seat kilometers (ASK), seats per operation, and route distance measured in both kilometers and nautical miles. Together, these variables provide valuable operational insight into the factors that influence how frequently airlines schedule service for individual markets.

Prior to model development, the dataset was examined to assess overall quality and determine whether preprocessing would be required. A duplicate record analysis confirmed that no duplicate observations were present within the dataset. Missing value analysis identified only 232 incomplete observations, representing less than 0.1% of the

total dataset. Because the percentage of missing values was extremely small and limited to the Operating Airline ASK and Distance variables, the incomplete observations were removed without materially affecting the overall analysis.

Following data cleaning, the final modeling dataset contained 250,786 observations and 21 variables, providing a large and reliable dataset, perfect for predictive modeling.

Data Preparation

After confirming the overall quality of the data, several preprocessing steps were performed to prepare the dataset for machine learning. First, observations containing missing values were removed to ensure that each record contained complete information for all predictor variables. Next, variables that duplicated existing information or were not required for model development were excluded to reduce redundancy within the dataset.

Several variables contained categorical information, including airport identifiers, operating airline, aircraft fleet, manufacturer, and schedule type. Because machine learning algorithms require numerical input, these categorical variables were encoded using LabelEncoder from the Scikit-learn library (Pedregosa et al., 2011). Encoding preserved each category while converting the values into numerical values that could be processed by the predictive models.

The cleaned dataset was then divided into predictor variables (X) and the target variable (y), with Departure Count serving as the target. An 80/20 train-test split was subsequently applied, resulting in 200,628 observations for model training and 50,158 for independent model evaluation. Separating the data this way ensures that model performance is evaluated using previously unseen observations and provides a more realistic assessment of predictive capability.

Preparing the data this way established a consistent foundation for both exploratory analysis and machine learning. Before predictive models were developed, exploratory data analysis was performed to better understand the dataset's characteristics and identify relationships between variables that could influence airline route demand.

Methods

The project followed a structured applied data science workflow consisting of exploratory data analysis, predictive modeling, model evaluation, and interpretation of

results. This approach allowed the dataset to be examined from both descriptive and predictive perspectives while ensuring that each stage of the analysis informed the next.

The first stage involved Exploratory Data Analysis, where descriptive statistics and visualizations were used to better understand the data distribution and identify potential relationships between the predictor variables. Histograms, bar charts, trend plots, and a correlation matrix were developed to examine route characteristics, airline representation, seasonal scheduling patterns, airport markets, and numerical relationships between the variables. These visualizations provided valuable context before predictive models were developed and helped identify characteristics that could influence model performance.

Following exploratory analysis, two supervised machine learning algorithms were evaluated. Linear Regression was selected as the baseline model because of its simplicity, interpretability, and widespread use in predictive analytics. The model provided an initial benchmark against which more advanced algorithms could be compared.

To better capture the complex relationships present within airline scheduling data, a Random Forest Regression model was also developed. Random Forest is an ensemble learning algorithm that combines the predictions of multiple decision trees to model nonlinear relationships while reducing the likelihood of overfitting. Because airline scheduling decisions are influenced by numerous interacting operational variables, Random Forest provided an appropriate alternative to traditional linear modeling techniques.

Model performance was evaluated using four commonly accepted regression metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). Together, these metrics provide a comprehensive assessment of predictive accuracy by measuring average prediction error, the magnitude of larger errors, and the proportion of variation explained by each model.

Analysis

Exploratory Data Analysis

Exploratory Data Analysis was conducted to better understand the dataset's characteristics before developing predictive models. The purpose of this phase was to identify trends, distributions, and relationships between the variables and to verify that the data was appropriate for machine learning. In addition to supporting feature selection, the exploratory analysis provided valuable business insight into airline scheduling patterns throughout North America.

The first step was to examine the distribution of the target variable, Departure Count. Understanding the target variable is critical because the performance of any predictive model depends on accurately representing the data it seeks to forecast. Figure 1 illustrates the distribution of scheduled monthly departures across all airline routes in the dataset.

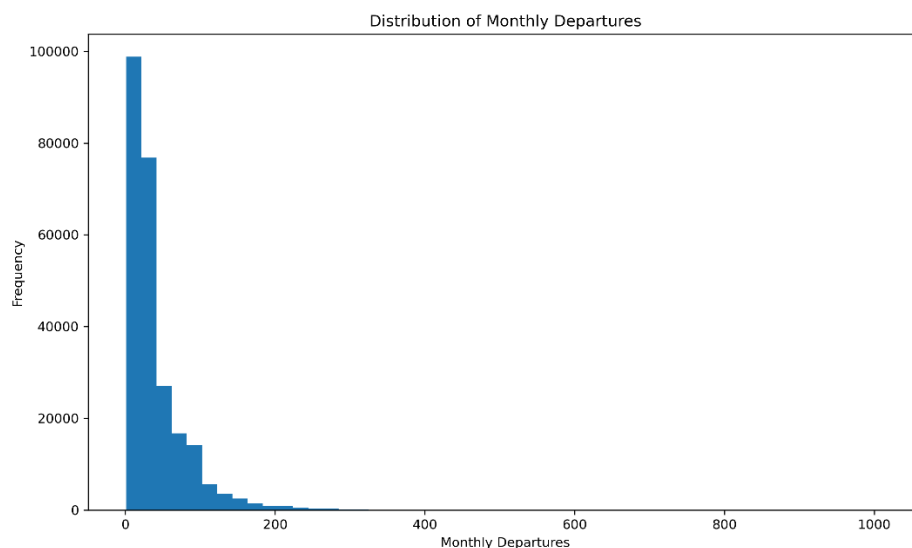


Figure 1. Distribution of Monthly Departures

As shown in Figure 1, the distribution is heavily right-skewed. Most airport markets operate relatively few departures each month, while a much smaller number of routes operate at significantly higher frequencies. This pattern is consistent with commercial airline operations, where only a limited number of major hub-to-hub markets receive exceptionally high service levels. The presence of this skewed distribution suggested that more flexible machine learning algorithms might outperform traditional linear methods.

After examining the target variable, the next step was to evaluate route distance, one of the dataset's primary operational characteristics. Figure 2 presents the distribution of route distances measured in nautical miles.

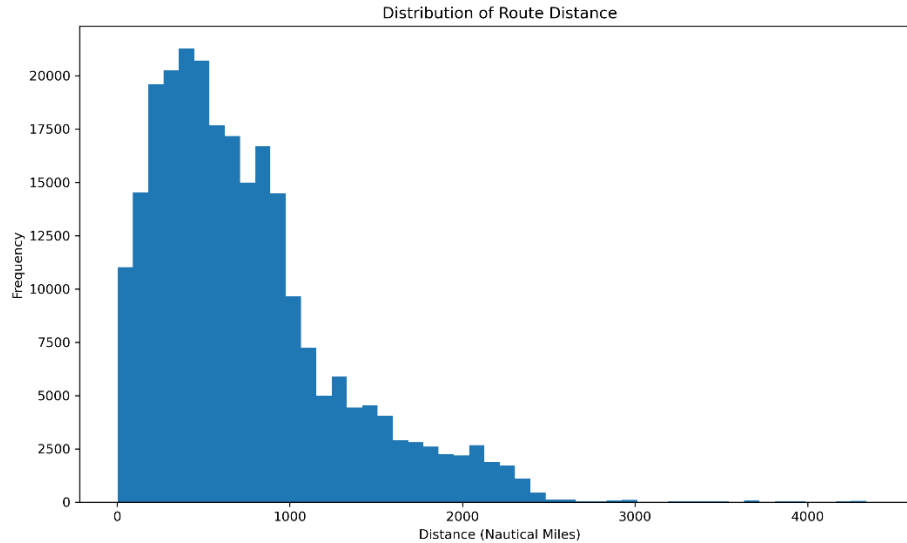


Figure 2. Distribution of Route Distance

Figure 2 shows that most scheduled routes fall within the short- to medium-haul range, with progressively fewer extending into long-haul operations. This distribution accurately reflects the structure of the North American airline network, where domestic and regional services account for most scheduled flights. Because aircraft selection, scheduling frequency, and operating costs are all influenced by route distance, this variable was expected to contribute meaningfully to the predictive models.

To better understand the dataset's composition, the operating airlines represented in the schedules were also examined. Figure 3 summarizes the 15 airlines with the most records.

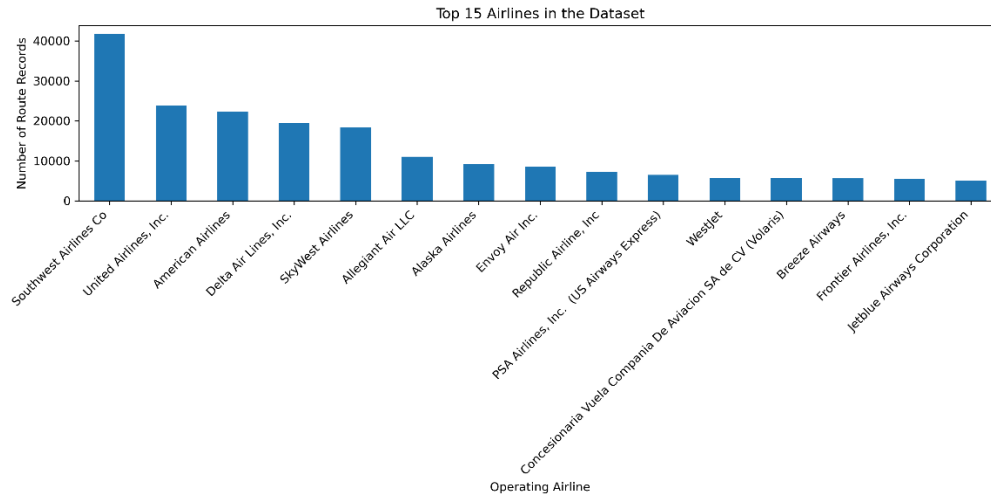


Figure 3. Top 15 Airlines in the Dataset

Southwest Airlines, United Airlines, American Airlines, Delta Air Lines, and SkyWest Airlines account for the largest share of observations within the dataset. These carriers operate extensive domestic and regional networks throughout North America, making their prominence within the dataset consistent with industry expectations. Examining airline representation also confirms that the dataset captures a broad cross-section of airline operations rather than reflecting a single carrier or business model.

The aircraft manufacturers listed in the schedules were then analyzed to better understand the study's fleet composition. Figure 4 summarizes the distribution of aircraft manufacturers.

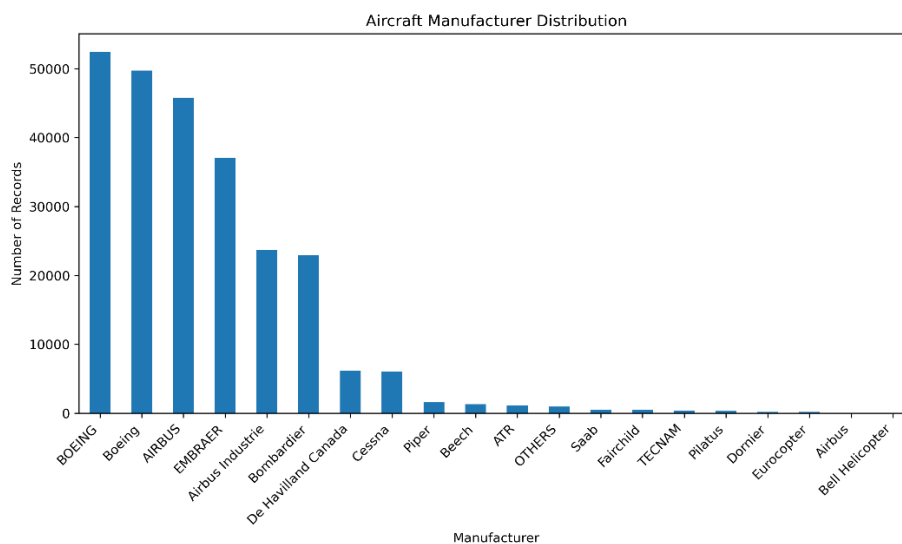


Figure 4. Aircraft Manufacturer Distribution

As expected, Boeing and Airbus aircraft account for the largest proportion of scheduled operations, followed by Embraer and several regional aircraft manufacturers. This distribution reflects the composition of commercial fleets currently operating across North America and further supports the dataset's representativeness.

Airport markets were also examined to identify the routes that appeared most frequently in the scheduling data. Figure 5 presents the 15 airport markets with the most scheduled observations.

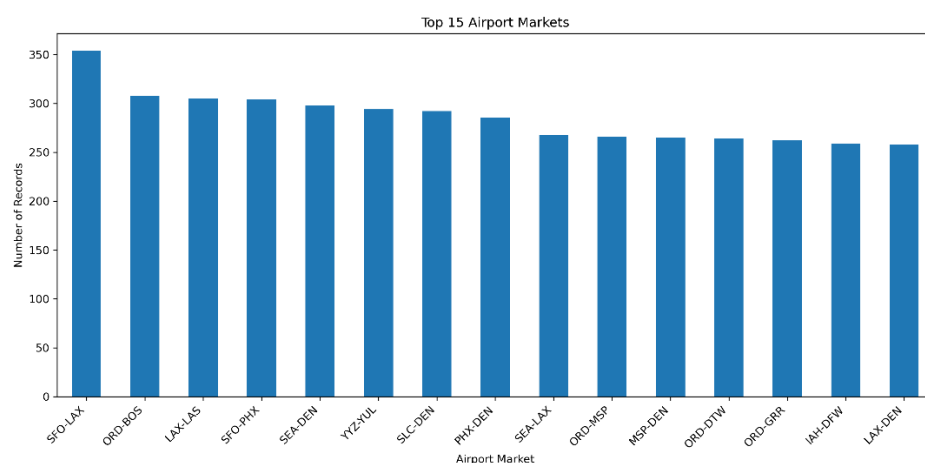


Figure 5. Top 15 Airport Markets

Several major domestic corridors appear among the busiest airport markets, including San Francisco–Los Angeles, Chicago O’Hare–Boston, and other high-frequency business and leisure routes. These markets represent locations where airlines regularly evaluate schedule adjustments, making them particularly relevant to demand forecasting applications.

In addition to examining route-level characteristics, average monthly departures were analyzed to identify potential seasonal scheduling patterns. Figure 6 summarizes the average departure counts by month.

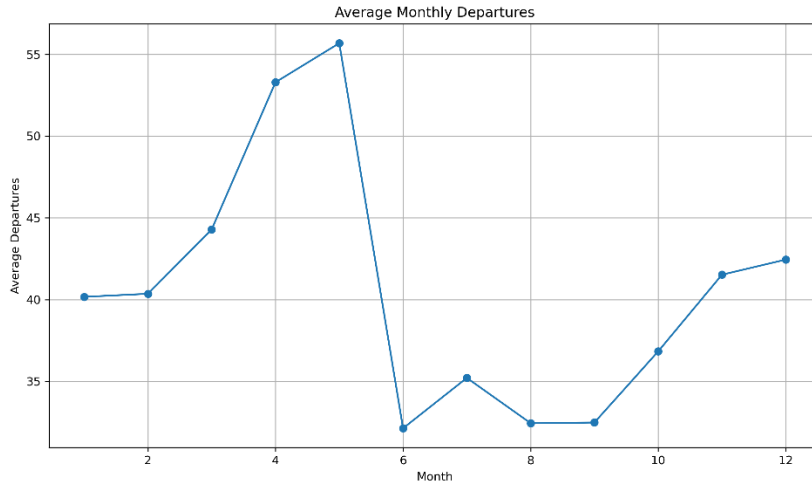


Figure 6. Average Monthly Departures

Although the monthly averages fluctuate throughout the year, the analysis suggests moderate seasonal variation in airline scheduling activity. These patterns indicate that the month may provide useful predictive information for estimating future departure frequencies and therefore should remain in the final feature set.

Finally, relationships among the numerical variables were evaluated using a correlation matrix. Correlation analysis provides an initial assessment of how strongly variables move together before more sophisticated predictive models are developed.

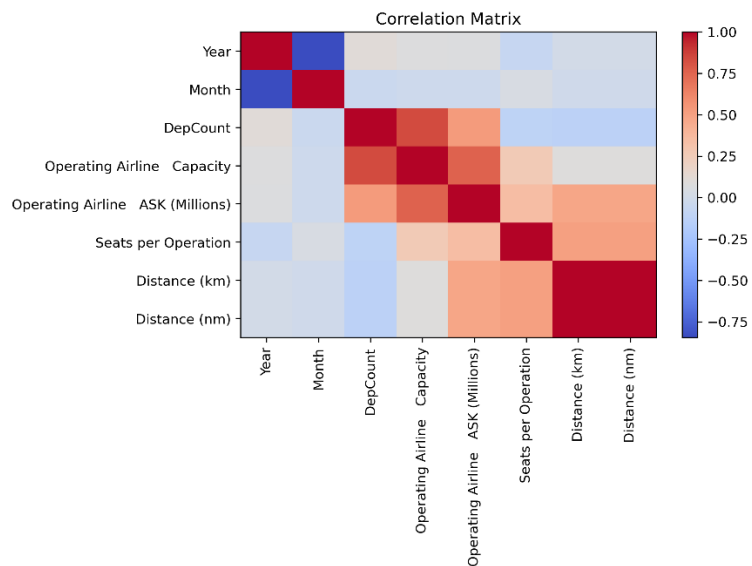


Figure 7. Correlation Matrix

The correlation matrix confirms several expected relationships within the dataset. Departure Count exhibits a strong positive relationship with airline capacity, while distance measured in kilometers and nautical miles is nearly perfectly correlated because both variables describe the same operational characteristic using different measurement units. Although correlation analysis identifies only linear relationships, these findings provided valuable insight into the dataset.

The exploratory analysis demonstrated that the dataset contains meaningful operational patterns suitable for predictive modeling. Building on these findings, the next phase of the project focused on developing machine learning models to predict monthly airline departures using the operational characteristics identified in the exploratory analysis.

Linear Regression

Following the exploratory analysis, a Linear Regression model was developed to establish a baseline for predicting scheduled monthly departures. Linear Regression was selected because it is one of the most widely used supervised learning algorithms for continuous prediction problems and provides a straightforward benchmark against which more sophisticated models can be evaluated.

The model was trained using the 80% training dataset and evaluated using the remaining 20% testing dataset to measure predictive performance on previously unseen observations. Performance was assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Table 2. Linear Regression Performance

Metric	Value
Mean Absolute Error (MAE)	8.97
Mean Squared Error (MSE)	281.48
Root Mean Squared Error (RMSE)	16.78
R^2	0.8545

Table 2. Linear Regression Performance

Table 2 summarizes the evaluation metrics for the baseline Linear Regression model.

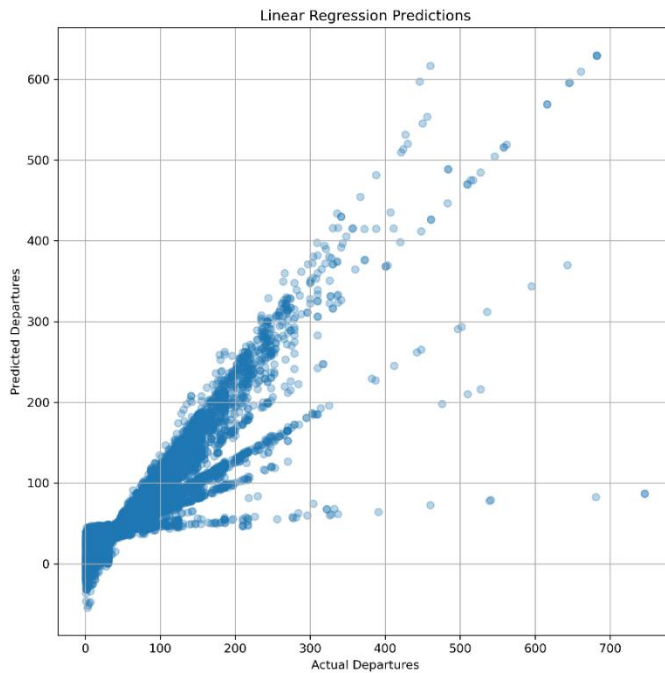


Figure 8. Linear Regression Predictions

Figure 8 compares the actual monthly departure counts with the values predicted by the Linear Regression model. While the model successfully captures the overall relationship between the predictor variables and the target variable, the dispersion of points increases as departure counts become larger. This indicates that the model struggles to accurately predict high-frequency routes, suggesting that the relationships within the dataset are more complex than a purely linear model can represent.

Although Linear Regression achieved an R^2 of 0.8545, providing a strong baseline for comparison, the results suggested that a more flexible machine learning algorithm could improve predictive performance by capturing nonlinear relationships between the operational variables.

Random Forest Regression

To better model the complexity of airline scheduling behavior, a Random Forest Regression model was developed. Unlike Linear Regression, Random Forest combines the

predictions of numerous decision trees, allowing it to capture nonlinear relationships and interactions among variables that are common in operational datasets (Breiman, 2001).

Using the same training and testing datasets, the initial Random Forest model produced exceptionally strong evaluation metrics.

Table 3. Initial Random Forest Performance

Metric	Value
Mean Absolute Error (MAE)	0.05
Mean Squared Error (MSE)	0.37
Root Mean Squared Error (RMSE)	0.61
R ²	0.9998

Table 3. Initial Random Forest Performance

Table 3 summarizes the performance of the initial Random Forest model.

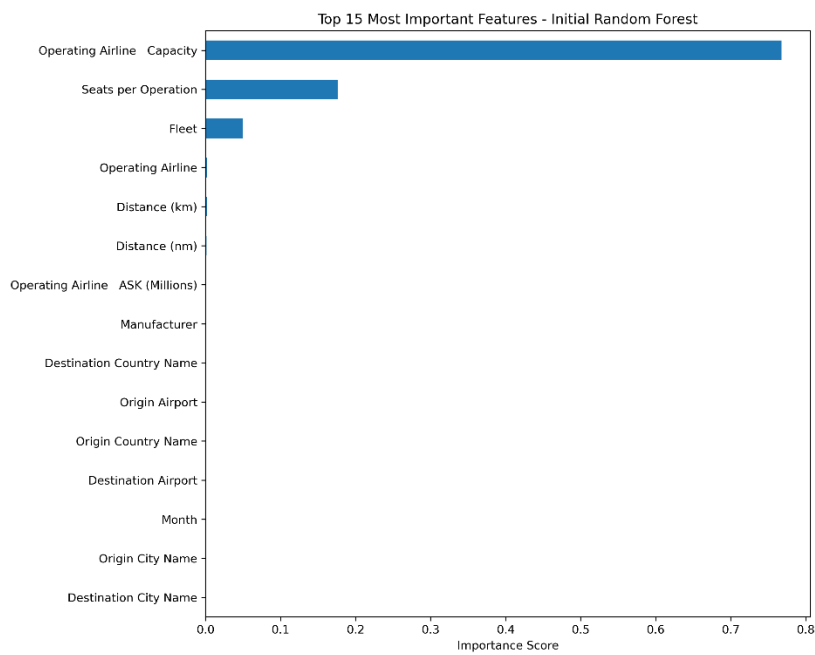


Figure 9. Initial Random Forest Predictions

At first glance, the model appeared to perform almost perfectly. The predictions closely matched the observed departure counts, yielding an R^2 of 0.9998, an RMSE of 0.61, and an MAE of 0.05 departures. While these results would typically suggest an outstanding predictive model, they were inconsistent with my expectations for a real-world airline forecasting problem.

Rather than accepting these results at face value, I performed additional investigations to determine whether the model's performance accurately reflected its predictive capability.

Identifying Data Leakage

One of the most important aspects of predictive modeling is ensuring that the model is evaluated using only information that would realistically be available when making future predictions. During the review of the initial Random Forest model, its near-perfect performance suggested it may have had access to information directly related to the target variable.

Further examination of the predictor variables confirmed this concern. Variables such as Operating Airline Capacity and Operating Airline ASK are calculated from operational data closely tied to the number of scheduled departures. Because these variables indirectly contained information about the target variable, the model was effectively learning the answer rather than identifying true predictive relationships.

Rather than reporting these inflated results, the leakage variables were removed from the feature set before rebuilding the model. This ensured that the final model relied only on operational characteristics that would reasonably be available during airline schedule planning.

Identifying and correcting this issue represented a critical step in the project because it improved the credibility of the analysis and produced a model that more accurately reflects real-world forecasting performance.

Revised Random Forest Regression

After removing the leakage variables, the Random Forest model was retrained using the revised feature set. Although the revised model no longer benefited from variables directly related to the target variable, it continued to demonstrate strong predictive performance while providing a more realistic assessment of forecasting accuracy.

Table 4. Revised Random Forest Performance

Metric	Value
Mean Absolute Error (MAE)	9.71
Mean Squared Error (MSE)	256.31
Root Mean Squared Error (RMSE)	16.01
R^2	0.8675

Table 4. Revised Random Forest Performance

Table 4 summarizes the evaluation metrics for the revised Random Forest model.

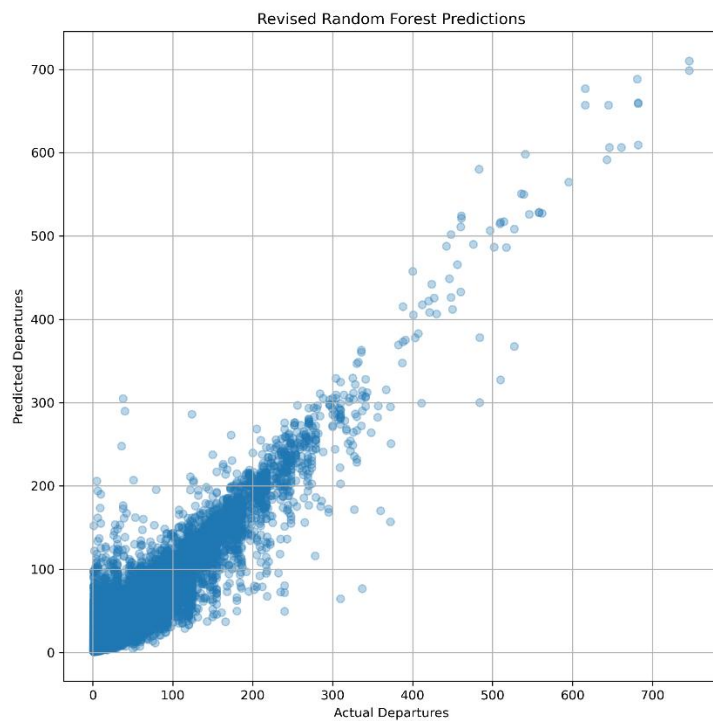


Figure 10. Revised Random Forest Predictions

Figure 10 shows that the revised model continues to follow the overall trend between actual and predicted departure counts, but exhibits greater variability than the initial model. This increase in prediction error is expected because the model no longer has access to variables that artificially inflate performance. The revised model achieved an R^2

value of 0.8675, an RMSE of 16.01 departures, and an MAE of 9.71 departures, providing strong predictive performance while maintaining methodological integrity.

To better understand how the revised model generated its predictions, feature importance was examined.

Table 6. Revised Random Forest Feature Importance

Feature	Importance
Distance (nm)	0.1396
Distance (km)	0.1365
Operating Airline	0.1025
Origin Airport	0.0897
Seats per Operation	0.0888
Destination Airport	0.0879
Month	0.0843
Destination City Name	0.0729
Origin City Name	0.0719
Fleet	0.0716

Table 6. Revised Random Forest Feature Importance

Table 6 ranks the variables according to their contribution to the revised Random Forest model.

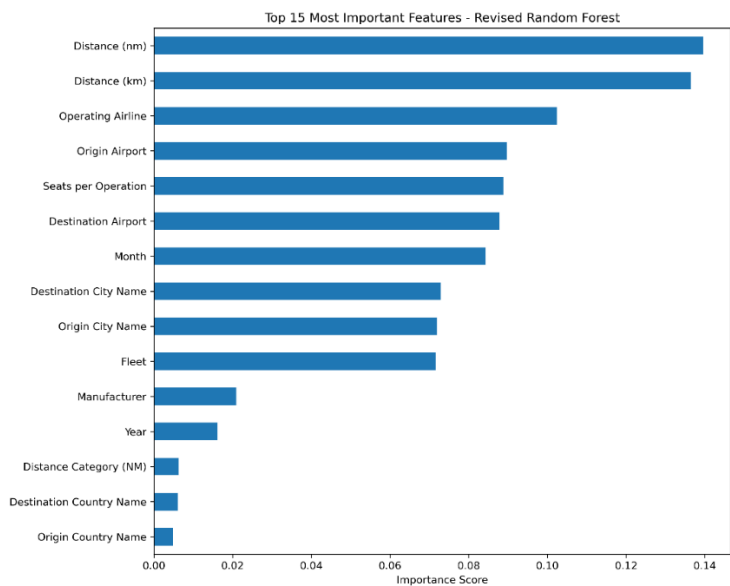


Figure 11. Revised Random Forest Feature Importance

The feature importance analysis indicates that route distance, operating airline, origin airport, destination airport, seats per operation, and month were among the most influential predictors of scheduled monthly departures. These findings align well with airline scheduling practices, where geographic characteristics, carrier strategy, aircraft capacity, and seasonal demand all influence flight frequency.

Model Comparison

The final step of the analysis compared the performance of all three predictive models.

Table 5. Model Comparison

Model	MAE	RMSE	R ²	Notes
Linear Regression	8.97	16.78	0.8545	Baseline model
Initial Random Forest	0.05	0.61	0.9998	Inflated by data leakage
Revised Random Forest	9.71	16.01	0.8675	Final selected model

Table 5. Model Comparison

Although the initial Random Forest model achieved the highest evaluation metrics, those results were invalid due to data leakage. After removing the leakage variables, the revised Random Forest remained the best-performing legitimate model, outperforming Linear Regression and providing realistic, trustworthy predictions. Consequently, the revised Random Forest model was selected as the final predictive model for this project.

The progression from a baseline Linear Regression model to an improved Random Forest model, followed by the identification and correction of data leakage, demonstrates the importance of rigorous model evaluation. Rather than selecting the model with the highest numerical performance alone, this project emphasizes the value of building models that generalize well to new data and accurately represent real-world forecasting conditions.

Conclusion

The objective of this project was to develop a machine learning model to predict monthly airline departures based on historical operational data from the SABRE Market Intelligence schedules database. Through a structured applied data science workflow, the project progressed from data preparation and exploratory analysis to predictive modeling, model evaluation, and interpretation of results.

The exploratory analysis demonstrated that the dataset accurately represents the diversity of North American airline operations. Variables such as route distance, airport market, operating airline, aircraft manufacturer, and seasonal scheduling patterns exhibited meaningful relationships with departure frequency, providing a strong foundation for predictive modeling.

Three predictive models were evaluated throughout the analysis. Linear Regression established a strong baseline by explaining approximately 85% of the variation in scheduled departures. Although the initial Random Forest model appeared to produce nearly perfect predictions, further investigation revealed data leakage from variables containing information directly related to the target variable. Rather than accepting these inflated results, the leakage variables were removed, and the model was rebuilt using only legitimate predictive features. The revised Random Forest model achieved an R^2 of 0.8675, indicating that it accurately predicts airline route demand and provides a realistic assessment of model performance.

One of the most significant outcomes of this project was not simply identifying the highest-performing model but demonstrating the importance of rigorous model validation. The discovery and correction of data leakage ensured that the final model reflects real-world forecasting conditions rather than artificially inflated evaluation metrics. This process highlights the importance of critically evaluating model performance rather than relying solely on numerical accuracy measures.

Overall, the revised Random Forest Regression model was selected as the preferred predictive model because it provided the best balance between predictive accuracy, interpretability, and practical applicability. The results demonstrate that machine learning can effectively support airline scheduling decisions by identifying operational characteristics that influence route demand while providing decision-makers with realistic forecasting capabilities.

Assumptions

Several assumptions were made throughout this analysis.

- The SABRE Market Intelligence schedules dataset accurately represents scheduled airline operations during the study period.
- Historical scheduling patterns are informative for predicting future departure frequencies.
- Scheduled monthly departures serve as a reasonable proxy for airline route demand.
- Airline scheduling practices remain relatively consistent over the period represented by the dataset.
- The operational variables included in the dataset contain sufficient information to develop meaningful predictive models without requiring additional data sources.

Limitations

Although the predictive models performed well, limitations should be acknowledged. The main limitation for this analysis is that it relied exclusively on airline scheduling data and did not incorporate external factors such as passenger demand, airfare, weather conditions, airport congestion, fuel prices, economic indicators, or regulatory changes. These variables may influence airline scheduling decisions and could improve predictive performance if incorporated into future models.

Challenges

Several challenges were encountered during the development of this project.

The most significant challenge involved identifying and correcting data leakage within the initial Random Forest model. Removing these variables required rebuilding the model and reevaluating its performance, ultimately producing a more realistic and trustworthy forecasting model.

Another challenge involved preparing the large operational dataset, which contained more than 250,000 observations. While the data was relatively clean, preprocessing, feature encoding, and model training required careful organization to ensure consistent results throughout the analysis.

Recommendations

The results of this study show that machine learning can provide valuable decision support for airline network planning and scheduling. Airlines could use predictive models, such as the revised Random Forest Regression model, to estimate future departure frequencies, evaluate proposed schedule changes, and identify routes that require additional operational review.

Future research should expand the feature set by incorporating external variables such as passenger demand, fare information, weather conditions, fuel prices, airport congestion, and regional economic indicators. Including these additional variables may improve predictive accuracy while providing a more comprehensive representation of the factors influencing airline scheduling decisions.

Organizations implementing predictive models should also establish procedures to routinely evaluate model performance and screen predictor variables for potential data leakage. Maintaining strong data governance and model validation practices will help ensure that predictive models continue to produce reliable forecasts over time.

Implementation Plan

A practical implementation strategy for this predictive model would consist of the following steps:

1. Obtain updated airline schedule data on a recurring basis from the SABRE Market Intelligence database.
2. Apply the same preprocessing and feature engineering procedures developed during this project.
3. Retrain the revised Random Forest model using the updated data.
4. Validate model performance using newly observed schedule information before operational deployment.
5. Integrate the model into the airline network planning process as a decision-support tool rather than a fully automated scheduling system.
6. Periodically monitor prediction accuracy and retrain the model as airline scheduling patterns evolve.

Following this implementation approach would allow airlines to maintain an accurate forecasting model while adapting to changing market conditions and operational requirements.

Ethical Assessment

Although this project used publicly available operational scheduling data and did not include personally identifiable information, ethical considerations remain an important aspect of predictive analytics.

Machine learning models should be used to support, not replace, professional judgment during airline scheduling and network planning. Decisions regarding flight frequency affect passengers, airline employees, airport operations, and surrounding communities. Consequently, predictive models should complement existing planning processes rather than serve as the sole basis for operational decisions.

This project also demonstrates the ethical importance of transparent model validation. Identifying and correcting data leakage was essential to ensure that the reported evaluation metrics accurately reflected the model's true predictive capability. Reporting the initial Random Forest results without acknowledging the leakage would have overstated the model's effectiveness and potentially misled decision-makers.

Finally, analysts should clearly communicate model assumptions, limitations, and potential sources of bias whenever predictive analytics are used to support business decisions. Transparent reporting promotes responsible artificial intelligence practices while increasing confidence in the resulting recommendations.

References

- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- SABRE Corporation. (n.d.). SABRE Market Intelligence. <https://www.sabre.com>

Appendix A – Data Dictionary

Origin Airport - Three-letter IATA code identifying the departure airport for the scheduled route.

Destination Airport - Three-letter IATA code identifying the arrival airport for the scheduled route.

Operating Airline - Airline responsible for operating the scheduled flight.

Fleet - Aircraft model or fleet assigned to operate the route.

Distance km - Great-circle distance between the origin and destination airports, measured in kilometers.

Capacity - Total number of scheduled passenger seats available on the route during the month.

Seats per Operation - Average seating capacity offered per scheduled departure for the route.

DepCount - Total number of scheduled monthly departures for the airline route; response variable predicted by the machine learning model.

Appendix B – Audience Questions

1. Why did you choose airline route demand prediction as the focus of this project?
2. Why was the SABRE schedules dataset selected for this analysis?
3. Why was DepCount chosen as the target variable?
4. Why were Linear Regression and Random Forest Regression selected for comparison?
5. What indicated that the initial Random Forest model contained data leakage?
6. How did removing the leakage variables improve the final model?
7. Which variables had the greatest influence on predicting airline route demand?
8. How could airlines use this model to support network planning and scheduling decisions?
9. What limitations should be considered before implementing this model in a production environment?
10. If this project were expanded, what additional data sources or machine learning techniques would you explore to further improve prediction accuracy?